

KAMALRAJ S

GEN AI DEVELOPER | AI/ML

Chennai, India | +91 8778437808 | kamalraj0653@gmail.com
GitHub: github.com/KAMALRAJ-S | LinkedIn: linkedin.com/in/kamalraj2003 | Portfolio: kamalraj-s.github.io/Kamalraj-Portfolio

PROFESSIONAL SUMMARY

Python developer with hands-on experience designing and deploying Generative AI solutions, including a production RAG-based conversational agent built on Microsoft Copilot Studio and a LangChain-powered document Q&A application, alongside a TensorFlow/CNN deep learning model. Practical expertise across the GenAI development lifecycle — prompt engineering, LLM integration, vector embeddings and similarity search, and retrieval-augmented generation (RAG) pipelines — combined with a strong foundation in object-oriented programming, data structures, and algorithms. Experienced in following SDLC and Agile/Scrum practices, including Git-based version control, code reviews, debugging, and technical documentation. Comfortable owning AI features from design through deployment, collaborating with cross-functional teams, and translating technical GenAI work for both technical and non-technical stakeholders.

TECHNICAL SKILLS

Generative AI & Agentic Development: Conversational AI Agents, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LLM Integration (Groq-hosted LLaMA 3.3), Hugging Face Transformers, Vector Embeddings & Similarity Search, Microsoft Copilot Studio

Machine Learning / Deep Learning: Machine Learning, Deep Learning, CNN, TensorFlow, Computer Vision (OpenCV), Model Training & Evaluation

Programming Languages: Python, SQL, JavaScript

Data & APIs: Pandas, NumPy, Matplotlib, Statistical Data Analysis, REST API Integration, JSON, SQL Querying & Database Management

Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, Exception Handling, Software Design Principles

Development Practices: Git / Version Control (GitHub), Debugging & Troubleshooting, Code Reviews, Clean & Maintainable Code, Testing, Defect Resolution, Technical Documentation, SDLC, Agile/Scrum, Problem Solving, Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Apprentice Software Engineer | Equiniti India, Chennai, India Jul 2025 – Jun 2026

- Designed and deployed a conversational AI agent using Microsoft Copilot Studio, applying a retrieval-augmented generation (RAG) approach to ground responses in a curated enterprise knowledge base.
- Applied Generative AI and RAG concepts — document-based knowledge retrieval, prompt design, and model configuration — to build, test, and refine AI agent behavior end to end.
- Developed a Python automation tool that parses protected report files into clean, structured data, reducing a 3–5 hour manual process to minutes and automating 60–70% of the workflow.
- Followed SDLC best practices including Git-based version control, code reviews, technical documentation, and requirements gathering in an Agile/Scrum environment, collaborating with cross-functional teams to understand requirements and troubleshoot issues.
- Participated in problem-solving and code review discussions with peers to maintain code quality and support defect resolution.

UI/UX Design Intern | KNILA IT Solutions, Coimbatore, India Feb 2024 – Mar 2024

- Designed wireframes and interactive prototypes for web and mobile applications in Figma, collaborating with developers to gather requirements and iterate on interface designs.

PROJECTS

RAG Document Assistant | Python, LangChain, Streamlit, ChromaDB, Hugging Face Transformers

- Built and deployed an end-to-end GenAI Python application enabling document upload and natural-language Q&A across PDF, DOCX, CSV, XLSX, and TXT formats via a Streamlit interface.
- Implemented embeddings, vector similarity search, Top-K retrieval, and LangChain prompt chains using Hugging Face Transformers to generate context-aware responses from a Groq-hosted LLaMA 3.3 model.

Diabetic Retinopathy Prognosis System | Python, TensorFlow, CNN, OpenCV, NumPy, Pandas

- Built a CNN-based deep learning model in TensorFlow to classify stages of diabetic retinopathy from retinal images, applying OpenCV-based preprocessing and NumPy/Pandas for dataset preparation and analysis.
- Developed and evaluated the end-to-end model training and classification pipeline.

EDUCATION

B.E. — Computer Science and Engineering

K. Ramakrishnan College of Technology, Tamil Nadu | 2020 – 2024

CERTIFICATION

Python Full Stack Development & Web Technology — QSpiders (Python, SQL, HTML5, CSS3, JavaScript)